

Mechanism Design Meets Contract Negotiation: Multidimensional Bilateral Trade with LLM Agents

Q Approximately Efficient Multidimensional Bilateral Trade [arXiv:2609.02872](#)

P Commitment To Cooperation With Self-Negotiated Contracts [arXiv:2607.22750](#)

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE

The Pair

Q studies multidimensional bilateral trade through mechanism design, providing simple mechanisms that achieve constant-factor approximation to first-best gains-from-trade (GFT) under Bayesian Incentive Compatibility (BIC), Interim Individual Rationality (IIR), and ex-ante Weak Budget Balance (WBB) [\[ledger 3\]](#). P examines whether LLM agents negotiating self-enforced contracts can improve cooperative outcomes in spatial-temporal environments [\[ledger 1\]](#).

Connection Pursued

Initial hypothesis: LLM agents using P's contract framework could empirically achieve Q's theoretical GFT guarantees. This was rejected as a category error [\[ledger 1\]](#).

Reframed hypothesis: Can LLM contract negotiation *emerge* mechanism-like behavior that approximates Q's theoretical bounds, rather than implementing Q's mechanism directly [\[ledger 4\]](#)?

Objections

The peers identified a fundamental architectural mismatch:

- Q uses central mechanism design: agents report types, mechanism computes allocation/payments via fixed rules [\[ledger 5\]](#).
- P uses decentralized contract negotiation: agents bargain over terms, then contracts are enforced [\[ledger 5\]](#).
- P's framework does not claim infrastructure for implementing arbitrary mechanism designs [\[ledger 2\]](#).

- Q assumes independent private values and analyzes equilibrium under truthfulness; P tests bounded-rational LLM behavior without mechanism constraints [\[ledger 1\]](#).

Supported Findings

The reframed direction is empirically testable [\[ledger 4, 5\]](#):

- Build bilateral trade environment with XOS buyer and additive seller valuations (from Q).
- Let LLM agents negotiate contracts using P's framework.
- Measure GFT versus Q's theoretical constant-factor bound.
- Analyze what contract structures emerge and whether enforcement enables outcomes closer to first-best than unenforced negotiation [\[ledger 4\]](#).

Prediction: if LLMs discover mechanism-like terms (e.g., "report values, split surplus per rule X"), contracts should approximate Q's bounds; if they bargain directly over trades, outcomes may exceed or fall short depending on negotiation skill [\[ledger 5\]](#).

Unresolved Issues

Evidence remains thin on:

- Whether P's contract representation language can express mechanism-like allocation rules.
- How to operationalize XOS valuations and additive costs in an environment LLMs can navigate (analogous to P's CT game but with trade structure) [\[ledger 3\]](#).
- Whether contract flexibility compensates for lack of incentive guarantees [\[ledger 3\]](#).

Smallest Next Step

Design a minimal bilateral trade environment combining Q's valuation structures with P's contract negotiation protocol. One XOS buyer, one additive seller, single trading round. Measure whether LLM-negotiated contracts achieve positive GFT and what terms emerge [\[ledger 4\]](#).